

# **BINF 6210/8210: Machine Learning for Bioinformatics**

## **Midterm Review**

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# Midterm Review: Topics

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- ▶ Data preprocessing
- ▶ Linear regression
- ▶ Classification
  - ▶ Logistic regression
  - ▶ Classifier evaluation

## Focus of the Review

The goal is not simply to memorize formulas, but to understand **what each method does, why we use it, and how to interpret it.**

## What Should You Be Able to Do?

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By the midterm, you should be able to:

- ▶ Explain concepts in your own words
- ▶ Interpret model parameters and outputs
- ▶ Compare closely related methods
- ▶ Identify inappropriate analysis strategies
- ▶ Explain **why** a method is appropriate or inappropriate
- ▶ Apply concepts to simple biological examples

# How Should You Review a Concept?

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For each concept, ask yourself:

What is it?



Why do we use it?



How does it work?



How do I interpret it?



What can go wrong?

# Data Preprocessing

# Why Preprocess Data?

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Raw biological data may contain:

- ▶ Different measurement scales
- ▶ Skewed distributions
- ▶ Outliers
- ▶ Missing values
- ▶ Systematic differences among samples
- ▶ Variables with very different variances

## Goal

Transform the data into a form appropriate for downstream analysis and modeling.

## Four Concepts You Must Distinguish

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| Operation             | Main Purpose                                            |
|-----------------------|---------------------------------------------------------|
| <b>Centering</b>      | Shift a variable's location                             |
| <b>Scaling</b>        | Change the relative magnitude or weighting of variables |
| <b>Transformation</b> | Change distributional shape                             |
| <b>Normalization</b>  | Reduce systematic differences among samples             |

### Important

Scaling and normalization address conceptually different problems.

# Centering

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For a variable  $x$ :

$$x' = x - \bar{x}$$

After centering:

$$\bar{x}' = 0$$

**What changes?**

- ▶ The location of the variable

**What does not necessarily change?**

- ▶ Variance
- ▶ Distributional shape
- ▶ Relative distances among observations

# Autoscaling

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Autoscaling:

$$z = \frac{x - \bar{x}}{s}$$

where  $s$  is the standard deviation.

After autoscaling:

$$\text{Mean} = 0$$

$$\text{Standard deviation} = 1$$

## Conceptual Effect

Variables originally measured on very different scales become more comparable.

# Why Can Autoscaling Be Problematic?

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Suppose:

- ▶ **Biomarker A**: large biological variation
- ▶ **Biomarker B**: almost constant, with small measurement noise

Autoscaling forces both variables to have:

$$SD = 1$$

If Biomarker B has a very small original standard deviation, dividing by that small value can magnify its small fluctuations.

## Key Idea

Equal variance after scaling does not necessarily mean equal biological importance.

# Autoscaling vs. Pareto Scaling

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## Autoscaling

$$x' = \frac{x - \bar{x}}{s}$$

## Pareto scaling

$$x' = \frac{x - \bar{x}}{\sqrt{s}}$$

- ▶ Autoscaling removes differences in variance more strongly.
- ▶ Pareto scaling preserves more of the original variance structure.

# Scaling vs. Normalization

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## Problem A

Metabolite concentrations range from 0.01 to 100,000.



Primarily a **variable-scale problem**

Think: Scaling

## Problem B

One sample has systematically higher signals because more material was injected.



Primarily a **sample-level systematic difference**

Think: Normalization

# Linear Regression

# Simple Linear Regression

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$$Y = \beta_0 + \beta_1 X + \epsilon$$

## Intercept

$\beta_0$  is the expected value of  $Y$  when  $X = 0$ .

## Slope

$\beta_1$  is the expected change in  $Y$  associated with a one-unit increase in  $X$ .

The model describes the **conditional mean of  $Y$** .

# What Does Ordinary Least Squares Do?

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For observation  $i$ , the residual is:

$$e_i = y_i - \hat{y}_i$$

Ordinary least squares chooses the coefficients that minimize:

$$\sum_i e_i^2$$

## Conceptual Interpretation

Find the regression model that minimizes the total squared prediction errors on the training observations.

# Multiple Linear Regression

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$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$$

Interpretation of  $\beta_j$ :

The expected change in  $Y$  associated with a one-unit increase in  $X_j$ , **holding all other predictors constant.**

**Important**

“Holding all other predictors constant” is an essential part of the interpretation.

# Why Can Regression Coefficients Change?

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Consider a simple model:

$$Y = \beta_0 + \beta_1 X_1 + \epsilon$$

Now add another predictor:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

The interpretation of  $\beta_1$  has changed.

In the multiple regression model,  $\beta_1$  asks:

How is  $X_1$  associated with  $Y$  among observations having the same value of  $X_2$ ?

# Confounding

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Suppose age is associated with both exercise and blood pressure:

$$\text{Exercise} \longleftrightarrow \text{Age} \longrightarrow \text{Blood Pressure}$$

A simple regression:

$$\text{Blood Pressure} = \beta_0 + \beta_1(\text{Exercise}) + \epsilon$$

may partly reflect age.

After adding age:

$$\text{Blood Pressure} = \beta_0 + \beta_1(\text{Exercise}) + \beta_2(\text{Age}) + \epsilon$$

the exercise coefficient may change substantially.

## Association Is Not Causation

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A regression coefficient describes an **association**.

It does not automatically answer:

*What would happen if we intervened and changed  $X$ ?*

Potential issues include:

- ▶ Confounding
- ▶ Omitted variables
- ▶ Selection effects
- ▶ Study design

# Logistic Regression

## Why Not Use Ordinary Linear Regression?

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For a binary outcome:

$$Y \in \{0, 1\}$$

we want:

$$0 < P(Y = 1 \mid X) < 1$$

A linear function can produce values below 0 or above 1.

Logistic regression uses the sigmoid function to map any real-valued linear predictor into the interval:

$$(0, 1)$$

# Probability, Odds, and Log-Odds

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## Probability

$$p = P(Y = 1)$$

## Odds

$$\frac{p}{1-p}$$

## Log-odds

$$\log\left(\frac{p}{1-p}\right)$$

| Probability | Odds  | Log-odds |
|-------------|-------|----------|
| $p = 0.5$   | $= 1$ | $= 0$    |
| $p > 0.5$   | $> 1$ | $> 0$    |
| $p < 0.5$   | $< 1$ | $< 0$    |

# Logistic Regression Is Linear in Log-Odds

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$$\log \left( \frac{p}{1-p} \right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

## Important

Logistic regression is **not linear in probability**.

It is linear in **log-odds**.

## Interpreting a Logistic Regression Coefficient

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Consider:

$$\log \left( \frac{p}{1-p} \right) = \beta_0 + \beta_1 X$$

For a one-unit increase in  $X$ :

log-odds change by  $\beta_1$

Exponentiating gives:

|                                      |
|--------------------------------------|
| Odds are multiplied by $e^{\beta_1}$ |
|--------------------------------------|

Thus:

$$e^{\beta_1}$$

is the **odds ratio** associated with a one-unit increase in  $X$ .

# Odds Ratio Is NOT Probability Ratio

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Suppose:

$$e^{\beta_1} = 2$$

Correct

A one-unit increase in  $X$  multiplies the **odds** of  $Y = 1$  by 2.

Incorrect

A one-unit increase in  $X$  doubles the **probability** of  $Y = 1$ .

Odds and probability are related by:

$$p = \frac{\text{odds}}{1 + \text{odds}}$$

Therefore, doubling the odds does not generally double the probability.

# From Probability to Class Label

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Logistic regression produces:

$$\hat{p} = P(Y = 1 \mid X)$$

To produce a class label, choose a threshold  $t$ :

$$\hat{Y} = \begin{cases} 1, & \hat{p} \geq t \\ 0, & \hat{p} < t \end{cases}$$

For example, if  $t = 0.5$ :

$$\hat{p} = 0.72 \Rightarrow \hat{Y} = 1$$

$$\hat{p} = 0.31 \Rightarrow \hat{Y} = 0$$

**Important**

Probability prediction and class prediction are not the same thing.

# Classifier Evaluation

## Start with the Confusion Matrix

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|          | Predicted 0 | Predicted 1 |
|----------|-------------|-------------|
| Actual 0 | TN          | FP          |
| Actual 1 | FN          | TP          |

Before memorizing a formula, ask:

- ▶ Are we starting with **actual positives**?
- ▶ Are we starting with **predicted positives**?
- ▶ Which type of error matters in the application?

## Four Core Metrics

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$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

# Recall vs. Precision

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## Recall / Sensitivity

Of all the **actual positives**, how many did we find?

$$\text{Recall} = \frac{TP}{TP + FN}$$

## Precision

Of all the **predicted positives**, how many were actually positive?

$$\text{Precision} = \frac{TP}{TP + FP}$$

## Accuracy Can Be Misleading

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Suppose we have 1,000 patients:

- ▶ 990 do not have the disease
- ▶ 10 have the disease

A classifier predicts **no disease for everyone**.

Then:

$$TN = 990, \quad TP = 0$$

Accuracy:

$$\frac{990}{1000} = 99\%$$

but:

$$\text{Recall} = 0$$

## What Happens When the Threshold Decreases?

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Suppose  $t : 0.5 \rightarrow 0.3$ :

More samples are classified as positive. Some actual positives may change:

$$FN \rightarrow TP$$

Therefore:

Recall generally increases

But some actual negatives may change:

$$TN \rightarrow FP$$

Therefore:

False positives generally increase

and precision may decrease.

# The Precision–Recall Tradeoff

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**Lower threshold**



More predicted positives



Usually:      Recall ↑      and Precision may ↓

**Higher threshold**



More selective about calling a sample positive



Usually:      Precision ↑      Recall ↓

# ROC Curve

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As the classification threshold changes, calculate:

$$TPR = \text{Sensitivity} = \text{Recall}$$

and

$$FPR = 1 - \text{Specificity}$$

Then plot:

|                |
|----------------|
| TPR versus FPR |
|----------------|

## Key Idea

Each point on the ROC curve corresponds to a particular classification threshold.

# Interpreting AUC

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AUC = Area Under the ROC Curve.

- ▶  $AUC = 1$ : perfect discrimination
- ▶  $AUC = 0.5$ : random discrimination
- ▶  $AUC < 0.5$ : ranking is effectively reversed

## Conceptual Interpretation

AUC is the probability that a randomly selected positive sample receives a higher model score than a randomly selected negative sample.

# Precision–Recall Curve

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Axes:

$$x = \text{Recall}$$

$$y = \text{Precision}$$

Precision–Recall analysis is especially informative when:

- ▶ The positive class is rare
- ▶ Identifying positive cases is the primary concern

## PR Focus

When the model says “positive,” how often is it correct, and how many true positives does it find?

# Overfitting and Regularization

# Recognizing Overfitting

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|                 | Training  | Unseen Data |
|-----------------|-----------|-------------|
| Underfitting    | Poor      | Poor        |
| Appropriate fit | Good      | Good        |
| Overfitting     | Very good | Poor        |

## Overfitting

The model learns patterns specific to the training data—including noise—rather than patterns that generalize to new data.

# Regularization

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Without regularization:

Minimize Loss

With regularization:

Minimize Loss +  $\lambda$ (Penalty)

## Conceptual Tradeoff

Accept a slightly worse fit to the training data in exchange for a simpler model with smaller coefficients that may generalize better.

# L1 vs. L2 Regularization

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|                                    | L1 / Lasso         | L2 / Ridge         |
|------------------------------------|--------------------|--------------------|
| Penalty                            | $\sum_j  \beta_j $ | $\sum_j \beta_j^2$ |
| Shrinks coefficients               | Yes                | Yes                |
| Can make coefficients exactly zero | <b>Yes</b>         | Generally no       |
| Implicit feature selection         | <b>Yes</b>         | No                 |

## What Does $C$ Mean in scikit-learn?

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For LogisticRegression:

$$C \propto \frac{1}{\text{regularization strength}}$$

Therefore:

small  $C \Rightarrow$  strong regularization

and

large  $C \Rightarrow$  weak regularization

**Question:**

Which model is more strongly regularized?

$$C = 0.01 \quad \text{or} \quad C = 100$$

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**Question:**

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$$C = 0.01$$

# **Generalization and Cross-Validation**

# Three Different Jobs for Data

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Training Data

Learn model parameters

Validation Data

Select models or hyperparameters

Test Data

Estimate final generalization performance

# The Test Set Has a Special Role

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The test set should:

- ▶ Remain untouched during model development
- ▶ Not determine  $C$
- ▶ Not determine which model is selected
- ▶ Not determine feature-selection decisions
- ▶ Be evaluated only after model development is complete

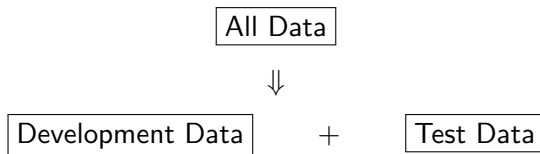
## Key Principle

Once the test set influences model selection, it is no longer an independent final evaluation set.

# Cross-Validation

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First split:



Perform  $K$ -fold cross-validation on the **development data only**.

- ▶ Train on  $K - 1$  folds
- ▶ Validate on the remaining fold
- ▶ Repeat until every fold has served as validation data
- ▶ Average validation performance

# What Is an Out-of-Fold Prediction?

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During  $K$ -fold cross-validation:

## Out-of-Fold Prediction

An out-of-fold (OOF) prediction is a prediction for a sample made by a model that was trained **without that sample**.

For example, in 5-fold CV:

Train on B+C+D+E

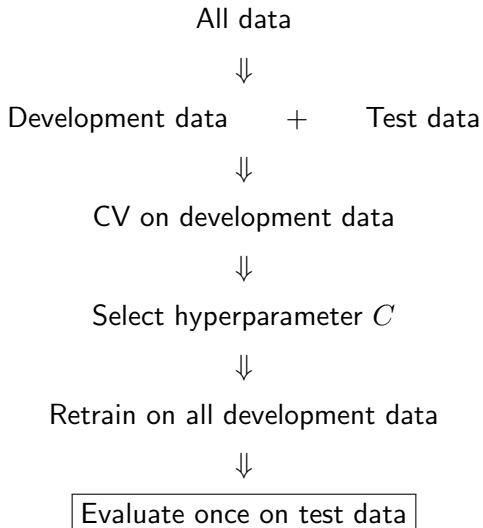


Predict samples in A

Those predictions for samples in A are out-of-fold predictions.

# Correct Hyperparameter-Selection Workflow

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# Concept Checks

## Concept Check 1

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A gene-expression variable has very small variance. After autoscaling, its small random fluctuations become prominent.

**Why?**

- A. Centering increases its variance
- B. Division by a small SD amplifies the fluctuations
- C. Normalization changes its biological meaning
- D. Logistic regression requires equal variance

## Concept Check 1

---

A gene-expression variable has very small variance. After autoscaling, its small random fluctuations become prominent.

**Why?**

- A. Centering increases its variance
- B. Division by a small SD amplifies the fluctuations
- C. Normalization changes its biological meaning
- D. Logistic regression requires equal variance

**Answer**

**B.** Division by a small SD amplifies the fluctuations.

## Concept Check 2

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A multiple linear regression gives:

$$\hat{Y} = 5 + 4X_1 - 3X_2$$

What does the coefficient 4 mean?

## Concept Check 2

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A multiple linear regression gives:

$$\hat{Y} = 5 + 4X_1 - 3X_2$$

What does the coefficient 4 mean?

### Answer

Holding  $X_2$  constant, a one-unit increase in  $X_1$  is associated with a 4-unit increase in predicted  $Y$ .

Which ideas are essential?

- ▶ **one-unit increase**
- ▶ **associated with**
- ▶ **holding  $X_2$  constant**

## Concept Check 3

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A logistic regression gives:

$$e^{\beta_1} = 0.5$$

What does this mean?

## Concept Check 3

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A logistic regression gives:

$$e^{\beta_1} = 0.5$$

What does this mean?

Correct

For each one-unit increase in  $X_1$ , the odds of class 1 are multiplied by 0.5.

Equivalent statement:

The odds are reduced by half.

Not Correct

The probability is reduced by half.

## Concept Check 4

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A classifier has:

Training accuracy = 98%

Validation accuracy = 70%

What should immediately concern you?

## Concept Check 4

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A classifier has:

Training accuracy = 98%

Validation accuracy = 70%

What should immediately concern you?

Overfitting

Possible approaches include:

- ▶ Increase regularization
- ▶ Reduce model complexity
- ▶ Reevaluate features or model design

## Concept Check 5

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A researcher tests 20 values of  $C$  on the **test dataset** and reports the best test accuracy.

**What is wrong?**

## Concept Check 5

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A researcher tests 20 values of  $C$  on the **test dataset** and reports the best test accuracy.

### What is wrong?

The test data have participated in:

Model selection

Therefore:

The final test estimate is no longer independent

Hyperparameters should instead be selected using validation data or cross-validation within the development data.

# Five Distinctions to Know Cold

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1. **Scaling vs. Normalization**
2. **Association vs. Causation**
3. **Probability vs. Odds**
4. **Precision vs. Recall**
5. **Validation Data vs. Test Data**

## A Useful Self-Test

If you can explain each pair clearly to another student **without looking at your notes**, you are well prepared for the conceptual portion of the midterm.

# Final Review Checklist

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Before the midterm, make sure you can explain:

- ▶ Why and when preprocessing is needed
- ▶ How to interpret linear regression coefficients
- ▶ Why association does not imply causation
- ▶ Probability, odds, and log-odd and how to interpret an odds ratio
- ▶ Accuracy, precision, recall, and specificity
- ▶ How classification thresholds affect performance
- ▶ ROC vs. Precision–Recall curves
- ▶ Overfitting and regularization, L1 vs. L2 regularization
- ▶ The meaning of  $C$
- ▶ Training vs. validation vs. test data
- ▶ How cross-validation is used for model selection

# Questions?

Good luck on the midterm!